Yuni Wu

South San Francisco, CA
yunihsuanwu@gmail.com | (720) 487-5403 | LinkedIn | GitHub | Portfolio

Summary

Robotics AI Engineer focused on making robot behavior measurable, reproducible, and deployment-aware. Builds Go1 telemetry pipelines, behavior-cloning policies, VLM navigation benchmarks, and safety-boundary evaluations that connect perception outputs to executable robot decisions.

Experience

Graduate Research Assistant Sept 2025 – May 2026
CAIRO Lab, University of Colorado Boulder — Robotics and embodied AI research lab

- Improved autonomous waypoint prediction to 0.11 MAE and $R^2 = 0.62$ by fine-tuning PyTorch/torchvision ResNet-18 behavior-cloning policies on structured 97K+ robot state-action samples.
- Converted ROS/rosbag telemetry into reproducible robot-learning datasets by aligning front-camera imagery, motion-command supervision, checkpoints, k-fold validation, and model-selection metrics.
- Validated physical Unitree Go1 inference behavior with ensembling, test-time augmentation, conservative smoothing, latency checks, stability validation, and safety monitoring.

Machine Learning Engineer Intern May 2025 – Aug 2025
MyEdMaster LLC — AI-driven healthcare and education technology company

- Verified 0.92 F1-score on a 22-class computer-vision workflow by evaluating ResNet-based models across 50K+ imbalanced samples.
- Reduced per-image inference latency by 5× (0.15s to 0.03s) by comparing ResNet architectures and measuring deployment tradeoffs.

Software Engineer June 2021 – Aug 2022
Shopee Co. Ltd. — Major Southeast Asian e-commerce platform

- Reduced manual QA workload by 80% by engineering Go API validation, SQL checks, and anomaly-detection workflows across 800+ production scenarios.

Projects

Multimodal Vision-Language Navigation for Unitree Go1 [project]

- Quantified VLM readiness for robot navigation by benchmarking Qwen3-VL-30B and InternVL-3.5-14B across 13 curated Go1 social-navigation rosbags with a five-label action interface.
- Measured uncertainty, directional activation, crossing detection, unsafe-forward behavior, latency, and bag-level consensus accuracy across 2.4K+ labeled state-action sequences.
- Evaluated slower VLM guidance as a high-level decision layer by building structured prompts, action parsing, controller-integration experiments, and safety-projected execution paths.

NailAI — Cloud-Native Vision Inference Platform [demo]

- Recorded 0 failures in a 200-concurrent-user load test with p95 latency of 6.8s through Cloud Run services, Pub/Sub job routing, idempotent orchestration, and retry handling.
- Instrumented request, job, and prediction metadata to monitor asynchronous image-inference behavior and downstream reliability signals.

Technical Skills

Languages: Python, SQL, Go, R, TypeScript
Robotics AI: ROS, rosbag, Unitree Go1, behavior cloning, waypoint prediction, VLM navigation evaluation, safety-projected control boundaries
ML & Evaluation: PyTorch, torchvision, NumPy, ResNet/CNNs, transfer learning, cross-validation, MAE, R^2, F1-score, calibration, latency checks
Systems: FastAPI, Docker, Cloud Run, Pub/Sub, Cloud Storage, BigQuery, metadata pipelines, monitoring, validation workflows, LangGraph, RAG, Kubernetes

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0 May 2026
Soochow University — BS, Data Science | GPA: 3.7/4.0 June 2021

Leadership

Taiwanese Student Association at CU Boulder, President 2025 – 2026
Coordinated student programming, cross-functional planning, and community events for Taiwanese students.